

Claims in Surfing: The Influence of Emotional Postperformance Expressions on Performance Evaluations

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The goal of the present research was to investigate whether claims (postperformance nonverbal emotional expressions) influence people in evaluating performance during surf contests. To test this research question, the authors sampled videos from professional surf contests and asked laypeople (Experiment 1; $N = 110$) and surf judges (Experiment 2; $N = 41$) to evaluate the performance in 2 online experiments. A subset of the surfing performances showed surfers displaying postperformance emotional expressions (claims), while another subset showed the same performances without the claims (nonverbal emotional expressions). Both experiments provided evidence that both laypeople and surf judges were biased by claims in judging surfing performances, with claims better than the performances without claims. The findings are in line with social-cognitive models emphasizing the socioconsequences of emotion expressions. The authors discuss the implications of the findings for sport competitions that rely on judging sport performance.

Keywords: emotion expression, nonverbal behavior, pride

Indeed, from many thousands of miles away, I could almost hear the judges double-take and reenter their scores when one of the world's best surfers throw their arms up in pseudo-joy and triumph. (https://www.surfer.com/features/king_of_the_claims/)

This quote published in *Surfer* magazine illustrates the topic of this study: Do claims (postperformance nonverbal emotional expressions) influence people in scoring waves during surf contests? Or, stated more generally, does individual postperformance nonverbal behavior influence observers' evaluation of the performance? Several relevant lines of research from different domains suggest that claiming (nonverbal celebration of performance in wave surfing) indeed has the potential to affect people evaluating the observed performance.

Claims can be regarded as an instance of emotion expression. Darwin (1872/1998) is often credited for launching the scientific study of emotion expressions with his seminal book *On the Expressions of the Emotions in Man and Animal*. Darwin's take on emotion expressions has led to the formulation of basic emotion theory (Ekman, 1992; Keltner, Tracy, Sauter, & Cowen, 2019) with the central tenet that humans have a set of distinct emotions that are autonomously (or automatically) expressed and understood in a universal manner. For example, humans are assumed to express happiness (Shiota et al., 2017) and/or pride (Tracy & Matsumoto, 2008) with universal nonverbal signals following pleasant experiences and successful achievements.

When considering nonverbal expressions in sports, and more specifically claims in surfing, nonverbal expressions are not necessarily automatic and, therefore, most likely do not result in the same universal expressions. Usually, athletes are exerting some

control over their nonverbal expression, particularly in instances of claiming a certain performance in surfing competitions. Hence, it is not surprising that professional surfers show different nonverbal behaviors when claiming waves, for example, raising a single fist in front of their chest, raising both arms above the head, or arms akimbo with hands on hips, etc. In this respect, Goffman (1959, 1963, 1971) was one of the first to explicitly state that people can purposely control their nonverbal expressions to convey particular impressions. Although Goffman's analyses were based on casual observations and anecdotal evidence, there is a vast amount of experimental evidence in the literature showing that nonverbal expressions can be used for self-presentation purposes (see DePaulo, 1992 for an early review).

Today, there is a general consensus that nonverbal expressions are under both conscious, deliberate control and unconscious, autonomous control (e.g., see Matsumoto, Frank, & Hwang, 2013 for a review). Hence, nonverbal expressions are conceptualized to systematically vary along a continuum of controllability (e.g., Ekman & Friesen, 1969). Although athletes have been shown to automatically display certain nonverbal behavior that conveys accurate information about internal states (Furley & Schweizer, 2014a, 2016a), people can also "fake" or modify certain nonverbal expressions in an attempt to influence observers. Claims in surfing can probably be considered a blend of spontaneous and intentionally used nonverbal expressions. In some instances, surfers will potentially claim a particular wave during a surf contest in an attempt to influence the judges, whereas in other cases, surfers might be so overwhelmed by the quality of a ridden wave that they will show an automatic claim.

In an attempt of providing a general model on the effects that emotional expressions can have on observers (e.g., on surf judges observing claims), Van Kleef (2009) proposed the *emotions as social information model* (EASI model) to better understand how emotional expressions may exert interpersonal effects. Central to the model is the assumption that emotional expressions affect observers via two routes: inferential processes and/or affective reactions. Inferential processes describe how an observer of the expression can infer certain information about internal states

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(e.g., feelings, attitudes, motivation, cognitions) of other people and the current situation. Applied to the present research, a judge or a spectator may conclude that a surfer has achieved something great (inference) when observing a display of pride (i.e., a claim in surfing), and should be evaluated accordingly (e.g., a higher score). On the other hand, the observed expressions can directly elicit affective reactions within the observer. One type of affective reaction may occur via the process of emotional contagion, whereby individuals catch the expresser's emotions through his or her facial expressions, bodily movements and postures, or vocalizations (Hatfield, Cacioppo, & Rapson, 1993). For example, when a friend starts laughing and one cannot help oneself and bursts out laughing too. According to the EASI model, this is more likely to occur among spectators (compared with judges) that support a particular surfer (because whether emotional contagion occurs or not depends on the relationship status of the expresser and the observer; Van Kleef, 2009). That means, people will likely experience similar emotional reactions when watching a video of their favorite team celebrating an important victory, whereas this is not likely to occur when watching similar footage of a team they do not support. Although inferential processes are a more likely candidate in explaining potential effects of surfing claims on performance evaluations, it is possible that in some instances claims might also be contagious to some degree on observers and thereby explain small portions of variance in resulting performance evaluations.

A further theoretical perspective on explaining how claims in surfing might influence observers' performance evaluations might be derived from a Brunswikian perspective (Brunswik, 1957) that has been an influential theoretical background in the context of sport officiating (Plessner, Schweizer, Brand, & O'Hare, 2009; Unkelbach & Memmert, 2010). More specifically, social judgment theory (Hammond, Stewart, Brehmer, & Steinmann, 1975) has transferred the theoretical concepts from Brunswik's Lens Model on perception to social situations such as officiating in sports. Regarding the potential influence of surfers' claims on performance evaluations, social judgment theory assumes that observers of surf contests try to judge a distal variable (i.e., surfing performance) based on multiple *proximal cues* (e.g., technical elements such as turns, fluency, barrels/tubes, speed, airs, etc.) that are correlated with the distal variable (i.e., surfing performance). Although it seems intuitive that *proximal cues* like certain technical elements will influence evaluations of surfing performance, it is less clear whether a claim after a surfing performance would also affect performance evaluations among observers. For this to be the case, observers would need to have learned (or believe) that the proximal cue (i.e., the claim) would have to be correlated with the distal cue (surfing performance). Given a related finding showing that soccer referees use crowd noise (Unkelbach & Memmert, 2010) as a proximal cue in judging the severity of fouls (i.e., awarding yellow cards) might indicate that postperformance emotional expressions in surfing would similarly be used by observers to inform performance evaluations, because both crowd noise in soccer and claims in surfing arguably often are valid cues in the respective evaluations. Hence, the present research tested if claims in surfing would be used as *proximal cues* by observers to inform (*distal*) performance evaluations.

In summary, there is a solid theoretical and empirical basis highlighting the potential influence of emotional expressions on observers. Therefore, the present studies attempted to empirically test the anecdotal observation highlighted in the initial quote from SURFER magazine if people in general (Experiment 1) and surf judges, in particular (Experiment 2), are influenced in their performance assessments by claims expressed by professional surfers.

The Present Research

Judgments of competitive performance are omnipresent in various sports. Although the outcome of some sport competitions can be determined by objective measurement (e.g., time in the 100-m sprint), an objective score (e.g., goals in hockey), several sport competitions rely on subjective judgments (e.g., points in gymnastics) to determine the outcome. This is also the case in competitive surfing. In professional surfing, athletes compete against each other in heats (depending on the contest either one-on-one, or heats containing three or four surfers, see <https://www.worldsurfleague.com/> for further details). Heats typically last 30–35 min (final heats sometimes 45 min and big wave heats 60 min) and surfers attempt to catch waves with high scoring potentials. Every wave an athlete surfs during a heat is given a score between 0.1 and 10 points. The scoring is done by a judging panel consisting of five judges, each declaring one score for a surfer's wave. The highest and lowest score gets eliminated and the mean of the remaining three scores determines the surfer's final wave score. The two best wave scores are summed at the end of the heat to comprise the surfer's final heat score (see WSL Rule Book 2019 [<https://www.worldsurfleague.com/asset/23142/2019+WSL+Rule+Book++03032019.pdf>] for more details).

We build on Plessner and Haar's (2006) social-cognitive perspective on sports performance judgments to investigate the influence of claims on evaluating surfing performance. This theoretical framework follows from the outlined Brunswikian perspective (Brunswik, 1957; Plessner et al., 2009) and suggests that observers of sport competitions integrate multiple cues (that are correlated or believed to be correlated with the to-be-judged performance) in constructing their performance evaluations that can lead to systematic biases in their judgments. Emotional expressions subsequent to surfing a wave during a contest can be considered a salient cue that observers of the performance integrate with their stored knowledge, the actual visual information from the surfing performance, crowd noise, etc., to construct a judgment of the witnessed performance (Plessner & Haar, 2006). Although, we argued that emotional expressions or claims can be modified or staged for self-presentational purposes (e.g., DePaulo, 1992), they can typically be regarded as correlated with surfing performance given that high-level surfers literally never claim a wave after a poor performance. In addition, the EASI model (Van Kleef, 2009) was built on a solid empirical basis demonstrating the strong impact of emotional expressions on observers. Therefore, we attempted to test the hypothesis that emotional expressions (claims) would lead to a systematic bias in judging surfing performance by both laypeople (Experiment 1) and experienced surfing judges (Experiment 2).

To test this hypothesis, we sampled two sets of stimuli published on the World Surf League (WSL; <https://www.worldsurfleague.com/>) event home pages for Experiments 1 and 2 showing video footage of professional surfers performing during contests. A subset of videos showed identical waves of surfers either claiming or not claiming the wave (i.e., the video was cut before the claim was shown). Given the problem of low-powered between-subject designs in experimental research in sport and exercise psychology (Schweizer & Furley, 2016), we decided to implement the crucial comparison of wave scores as a function of surfers either showing claims after performing or not showing a claim within subject. In order to have sufficient reliability, we averaged a subset of wave scores with claims and without claims (eight waves, respectively, in Experiment 1 and 10 waves, respectively, in Experiment 2) that were presented in randomized order among other waves to distract from the purpose of the experiments.

To ensure that any potential differences in wave scores were in fact due to observing claims versus no claims and not performance differences in the video stimuli, we randomly assigned participants to two groups. Group 1 presented one subset of waves that displayed claims after performance and another subset of waves that were cut before the claim was visible. Group 2 was shown the identical waves with the only difference that the same waves that were shown in Group 1 with claims were shown in Group 2 without claims and the waves shown without claims in Group 1 were shown with claims in Group 2. This methodology allowed us to test our central hypothesis that identical surf performances would be judged better if they were followed by a claim compared with the same video footage not showing the claim.

Experiment 1

Methods

Participants. A total of 110 participants (69 males, 41 females; $M = 24.9$; $SD = 4.4$) completed an online experiment. Sample size was calculated prior to the study to have sufficient power (0.95) to detect medium effects ($f = 0.2$) on the claim versus no claim ratings in a 2×2 analysis of variance (ANOVA; Group 1 vs. Group 2 $\times$ Claim vs. No Claim; [Faul, Erdfelder, Lang, & Buchner, 2007](#)).

Thirty-one participants (28.18%) reported having no experience in surfing. Among the other 79 participants (71.82%), 16.36% referred to themselves as beginners, 29.09% as intermediates, 24.56% as advanced, and 1.82% as experts regarding their level of surfing expertise. Furthermore, 31.82% reported having less than 1 year of experience in surfing, 47.27% between 1 and 5 years, and 20.0% more than 5 years; 40.91% had no visual experience in surfing, meaning not following a surfing event either on television, on the Internet, or at the beach prior to the study. The other 59.09% were following surf events on television, on the Internet, or at the beach where the event took place ($M_{\text{number of surf events/year}} = 2.0$; $SD = 2.43$). Of them, 72.6% were familiar with the international scoring system in surfing, whereas the other participants (27.4%) were not familiar with any type of judging criteria. Informed consent was obtained from every participant online before commencing the experiment.

Stimuli. The selected video sequences were sampled online at the video database Heat Analyzer (www.worldsurfleague.com). For the purpose of the present research, we sampled stimulus material from two competitions of the 2014 and 2015 World Championship Tour (Event Hurley Pro at Trestles 2014/2015; Quicksilver Pro Gold Coast Event 2015). The sampled videos always started with a surfer paddling for a wave and catching the wave, followed by the surfer stepping on the board and starting to surf the wave until the end (“the end” can have two different meanings—either the surfer finishes all his maneuvers successfully or the surfer falls).

Every video stimulus only displayed one surfer. Overall, 20 athletes were shown in this study. The length of the video sequences was determined by the performance of the surfer. The maximum length of one sequence lasted no longer than 40 s. The videos were divided into three categories: with claim, no claim, and unfinished wave. For the category “with claim,” waves were selected in which surfers showed claims (i.e., nonverbal emotional expressions) after finishing the wave (e.g., showing their fist [see Figure 1], raising arms above head, etc.). The claims occurred immediately after finishing the wave and had to be easily observable. For the category “no claim,” the same waves were used as in the category “with claim,” with the only exception that we cut the



Figure 1 — Sample frame taken from one of the experimental video stimuli used in Experiment 1 showing a surfer claiming a wave.

video before the claim was observable (i.e., only showing the surfing performance without the claim). For the category “unfinished wave,” only sequences were selected in which the surfer lost his balance and fell in the water. These waves were not used for further analyses, but instead served to distract participants from the purpose of the study.

Measure. Perceivers rated the short video scenes on a digital semantic differential scale after every video. In order to give their ratings, perceivers moved a mouse cursor from the middle of the scale that represented a medium score (50) toward either pole (0 = *lowest possible score*; 100 = *highest possible score*) of the scale and logged in their rating by clicking the left mouse button. The utilized scale was continuous.¹

Experimental Manipulation. Participants were randomly assigned to one of the two groups. Group 1 was shown a subset of eight waves (category “with claim”) that displayed claims after performance. Group 1 was further shown eight waves that were cut before the claim was visible (category “without claim”). Group 2 was shown the identical waves with the only difference that the same waves that were shown to Group 1 “with claims” were shown to Group 2 “without claims” and the waves shown “without claims” to Group 1 were shown “with claims” to Group 2. This manipulation enabled us to have a powerful test comparing evaluations of waves “with claim” and “without claims” within participants while controlling for differences in wave quality and surfing performance. In addition, we had three waves occurring twice—once with claim and once without—for every participant in both Groups 1 and 2 to compare these evaluations. The problem with this approach, however, might be that participants notice that they were evaluating the same waves twice (which we controlled for by asking participants after the experiment).

Procedure. Perceivers were instructed to judge the quality of surfing performance based on the video footage presented to them by moving a mouse cursor to either the “very low” or “very high” pole of the scale. They were further instructed to answer as accurately as possible, whereas speed was not emphasized. Before commencing the experiment, perceivers filled out an online questionnaire gathering demographic data. Each participant had to rate a total of 27 waves. These 27 videos included five “falling waves,”

eight waves with claim, and eight waves without claim. In addition, each participant saw three waves twice: once showing the wave with a claim and once with the same video cut before the claim was visible.

All videos were presented silently. The participants were first shown an example video sequence to familiarize them with the procedure. After using the mouse to click on a “forward button” on the screen, the participants were asked to judge the performance by the surfer using the scale presented previously. There was no time limit for the judgment. After confirming a judgment, participants were automatically forwarded to the next sequence, which then started automatically. After judging all 27 video sequences, the perceivers answered three control questions to check if the perceivers experienced any unexpected occurrences, if they realized that some sequences were presented multiple times, and finally they were asked if any problems occurred, while watching the videos.

Data Analyses

A significance level of $\alpha = .05$ was defined for the test. To test the hypothesis, a two-factor mixed-design ANOVA was conducted. The within-subject factor claim (claim vs. no claim) tested whether claimed waves were judged higher than unclaimed waves. The between-subjects factor group (Group 1 vs. Group 2) was conducted to exclude that the two experimental groups differed in their judgments. In addition, a t test for dependent samples was conducted to test if the subset of waves (three waves) that were presented twice to participants differed depending on whether it was seen with a claim or without a claim.

Results

The descriptive results are displayed in Figure 2. A mixed-design ANOVA with the within-subject factor claim (claim vs. no claim) and the between-subject factor group (Group 1 vs. Group 2) revealed a significant main effect of claims on overall performance judgments, $F(1, 108) = 9.175, p = .003, \eta_p^2 = .087$, indicating average higher performance ratings for video footage of waves that showed surfers claim waves as opposed to waves that were not claimed. There was no significant main effect of group on wave scores (the waves that were shown with claims to one group were shown without claims to the other group and vice versa), $F(1, 108) = 0.300, p = .585, \eta_p^2 = .003$, indicating that the two

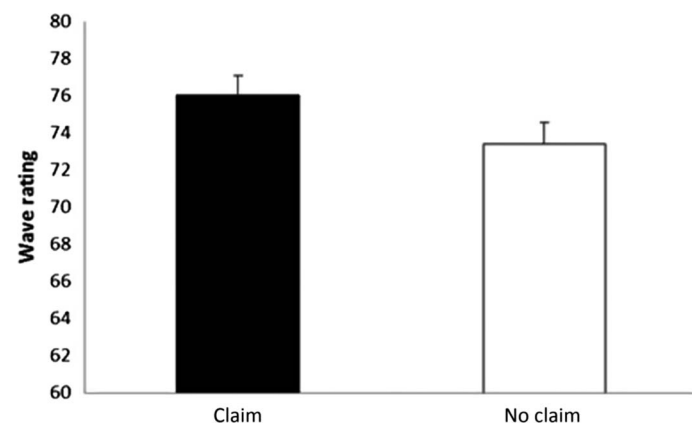


Figure 2 — Mean score (0 = lowest, 100 = highest) estimates as a function of whether or not the video showed a surfer claiming a wave. Error bars represent SEs.

groups judged the shown waves similarly overall. The interaction between claim and group was not significant, $F(1, 108) = 1.209, p = .274, \eta_p^2 = .011$.

When comparing the subset of waves (three waves) that were presented twice to participants, once with a claim and once without a claim, no significant mean difference emerged, $t(119) = -.540, p = .590$, two-tailed; $d = 0.05$. Potentially, this experimental manipulation was too obvious as participants were seeing the same waves twice (although only 36 out of 110 stated that they noticed that they were viewing some of the waves twice).

Discussion

Experiment 1 provided first evidence that people seem to be systematically biased to judge waves with claims better than waves without claims, or stated differently that observers seem to use claims as *proximal cue* to inform their (*distal*) performance evaluations. The effect size of this effect can be considered medium by convention. We did not find this bias in the three waves that were presented twice (once with claim and once without claim) to the same participants. Potentially, the experimental manipulation was too obvious as participants were seeing the same waves twice (although only 36 out of 110 stated that they noticed that they were viewing some of the waves twice).

Although it seems interesting that laypeople are biased in their performance judgments by postperformance emotional expressions, the more important question is if surf judges would also be biased by claims in their performance evaluations. As surf judges are trained to give as objective performance evaluations as possible, it is not clear if surf judges would also be influenced by claims in their performance evaluations. Therefore, Experiment 2 tested the same research question with a slightly modified experimental setup and a different participant population (i.e., surf judges).

Experiment 2: Surf Judges

Although the main rationale of Experiment 2 was to test if the effect found in Experiment 1 would also be observable in surf judges who are trained to give objective performance evaluations, Experiment 2 further followed a call by Fiedler (2011), who pointed out the necessity of replicating effects found with one set of stimuli with different stimuli to ensure that the phenomenon of interest does not only apply to a highly specific set of stimulus material, but also applies generally to the phenomenon of interest.

Methods

Participants. Forty-one participants (38 males, two females, and one diver; $M = 32.32$; $SD = 9.5$) completed the online survey. Fourteen participants (34.1%) had an International Surfing Association (ISA) Level 1 judging license, five participants (12.2%) had an ISA Level 2 judging license, nine participants (22.0%) had an ISA Level 3 judging license, and 13 participants (31.7%) had no judging license, but a sufficient amount of judging experience. The sample reported to have on average 36.05 ($SD = 84.46$) hr/year contest judging experience, and 9.44 ($SD = 18.2$) of competing experience hours/year of contest experience. Just as in Experiment 1, participants were randomly divided into two groups ($n = 21$ and $n = 20$, respectively). Informed consent was obtained from every participant via the experimental software before commencing the experiment. Given the limited availability to sample a sufficient number of surf judges to obtain similar power as in Experiment 1,

Experiment 2 was not as highly powered as Experiment 1. Nevertheless, a post hoc power analysis revealed that our sample had sufficient power (0.87) to detect medium effects ($f=0.25$) on the claim versus no claim ratings in a 2×2 ANOVA (Group 1 vs. Group 2 $\times$ Claim vs. No Claim; Faul et al., 2007).

Stimuli. Like in Experiment 1, the selected video sequences were sampled online on the WSL Heat Analyzer database, containing stimulus material from the World Championship Tour 2018 contest Corona Open J-Bay. Every sequence was muted and started just prior to the take-off and ended after finishing the performance (either by finishing all maneuvers or by falling). The length of the sequences varied between 3 and 35 s, depending on the performance of the surfers. Twenty-two surfers coming from eight different countries were presented to the participants. The sequences were divided into four categories: claim, no claim, completed, and wipe-out. The category “claim” contained videos where surfers claimed (i.e., nonverbal emotional expression) their surfed waves immediately after successfully completing the last maneuver. The category “no claim” contained the same performances as the “claim” videos with the only difference that the claims were cut out. Sequences from the category “completed” showed surfers completing the performance without claiming and the category “wipe-out” contained sequences where athletes fall off their boards. The last two categories were only shown to distract participants from the actual purpose of the study.

Measure. Perceivers rated the short video scenes on a digital semantic differential scale after every video. In order to give their ratings, perceivers moved a mouse cursor from the middle of the scale that represented a medium score (5) toward either pole (0.1 = lowest possible score; 10 = highest possible score) of the scale and logged in their rating by clicking the left mouse button. The utilized scale was continuous.

Procedure. Before the start of the experiment, participants filled out a questionnaire gathering demographic data. Perceivers were instructed to judge the quality of surfing performances according to official ISA judging criteria by moving a cursor either toward the “0.1 points” or the “10 points” pole of the scale. They were not informed about the actual goal of the experiment. Instead, they were instructed to answer as accurately as possible. Then, an example wave was shown to familiarize them with the procedure. Participants were instructed to give 6.5 points (utilizing the scale explained earlier) and to relate to this example when judging the following sequences. After clicking on a “continue” button on the screen, participants were randomly assigned to one of the experimental groups and the experiment started. Every participant got to see and judge 10 “claim” waves, 10 “no claim” waves, five “completed” waves, and five “wipe-out” waves. A “claim” stimuli and a “no claim” stimuli were always followed by one of the “filler” (completed or wipe-out) sequences. It was not possible to replay a stimulus. Every video in which one group got to watch a “claim” sequence was played to the other group as “no claim” version and vice versa. There was no time limit for the judgment. After confirming a judgment, participants were automatically forwarded to the next sequence, which then started automatically. After completing to judge all 30 video sequences, the participants had the chance to leave a comment.

Data Analyses

A significance level of $\alpha = .05$ was defined for the test. To test the hypothesis, a two-factor ANOVA was conducted. The inner

subject factor claim (claim vs. no claim) tested whether claimed waves were judged higher than unclaimed waves. The between-subject factor group (Group 1 vs. Group 2) was conducted to exclude that the two experimental groups differed in their judgments. To measure the influence of judge licenses, judging experience, and competition experience, a series of correlations were computed on the mean difference between claimed and non-claimed waves.

Results

The descriptive results are displayed in Figure 3. A mixed-design ANOVA with the within-subject factor claim (claim vs. no claim) and the between-subject factor group (Group 1 vs. Group 2) revealed a significant main effect of claims on overall performance judgments, $F(1, 39) = 16.478$, $p = .0001$, $\eta_p^2 = .297$, indicating average higher performance ratings for video footage of waves that showed a surfer claim waves as opposed to waves that were not claimed. There was no significant main effect of group on wave scores (the waves that were shown with claims to one group were shown without claims to the other group and vice versa), $F(1, 39) = 1.331$, $p = .256$, $\eta_p^2 = .033$, indicating that the two groups judged the shown waves similarly overall. The interaction between claim and group was not significant, $F(1, 39) = 1.1319$, $p = .258$, $\eta_p^2 = .033$.

To investigate if judging experience had an influence on whether claims had an impact on the scoring of judges, we correlated the ISA level ($p = -.078$, $p = .628$), judging experience ($r = .227$, $p = .154$), and contest experience ($r = -.023$, $p = .886$) with the mean difference between claimed waves and not claimed waves. The nonsignificant correlations indicate that all judges were similarly affected by the claims, regardless of their judging level and the amount of experience they had.

Discussion

Experiment 2 replicated the results from Experiment 1 with surf judges and different stimulus material. Hence, it provided evidence that surf judges also seem to be systematically biased to judge waves with claims better than waves without claims. Therefore, also experienced judges used claims as a *proximal cue* to inform their (*distal*) performance evaluations. The effect size of this effect

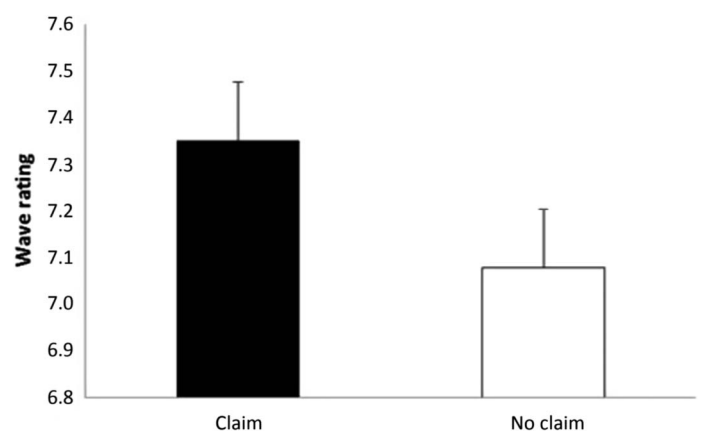


Figure 3 — Mean score (0=lowest, 10=highest) estimates as a function of whether or not the video showed a surfer claiming a wave. Error bars represent SEs.

can be considered large by convention (but see [Schweizer & Furley, 2016](#) on the danger of small[er] sample sizes on inflating effect sizes). Hence, Experiment 2 supported the anecdotal observation captured in the quote from SURFER magazine that claiming waves during surf competitions can influence the judges' performance evaluations. Although surf judges are trained to give as objective performance evaluations as possible, they appear to be systematically influenced by claims in their performance evaluations.

General Discussion

The central aim of the present research was to investigate if claiming (postperformance nonverbal emotional expressions) influences people in evaluating performance during surf contests. Both experiments provided evidence that both laypeople and surf judges were biased by claims in judging waves with claims better than waves without claims.

These findings are in line with Plessner and Haar's (2006) social-cognitive perspective on sports performance judgments and the outlined Brunswikian perspective of sports officiating ([Plessner et al., 2009](#)) as they provide experimental evidence that postperformance emotional expressions following surfing performance can be considered a salient cue that influences the information processing of observers and ultimately results in a systematic bias of performance judgments: that is, resulting in surf performance being judged more favorably if it is followed by a claim. As surfing competitions rely on subjective judgments of judges to determine the outcome, these findings have important applied implications for judges and training programs ought to implement interventions to avoid or reduce these biases. For example, video-based training ([Schweizer, Plessner, Kahlert, & Brand, 2011](#)) might help to sensitize surf judges toward claims and thereby help to diminish this bias.

In addition, the findings are in line with the more general *EASI model* ([Van Kleef, 2009](#)) by demonstrating that emotional expressions following athletic performance can have interpersonal effects. Although it seems more likely that the interpersonal effects were primarily caused by inferential processes, we cannot rule out that some of our participants experienced some emotional contagion when viewing the claims of the surfers. Most likely, differential performance evaluations between claimed waves and waves without claims were caused by participant's inferential processes: for example, they observed a display of pride and likely concluded that this surfer had achieved something extraordinary and therefore evaluated him accordingly. Nevertheless, we consider it a fruitful avenue for future research to illuminate the mechanisms of how observers are influenced by emotional expressions of athletes during sport competitions (see [Furley, Moll, & Memmert, 2015](#); [Moll, Jordet, & Pepping, 2010](#)).

In addition, we consider it interesting to combine the *EASI model* ([Van Kleef, 2009](#)) with social judgment theory ([Hammond et al., 1975](#)) or the multiple cue perspective of sports officiating ([Plessner et al., 2009](#)) to investigate into the relative importance of postperformance emotional expressions (e.g., claims in surfing) compared with other cues related to the to-be-judged performance. Currently, we are not aware of any research systematically researching the effects of emotional expressions on different kinds of performance evaluations. Given the importance of emotional expressions in social interactions (e.g., [Van Kleef, 2009](#)), it seems a feasible hypothesis that emotional expressions of athletes or other performers (e.g., musicians) might "trump" other cues that are

more relevant for accurate performance evaluations. For example, "a winning smile" of a figure skater or a dancer after a performance might influence judges more than certain technical elements of the performance as previous research outside the sports domain has indicated that "liking" can affect performance evaluations ([Cardy & Dobbins, 1986](#)). Future research in sports officiating and other domains involving performance evaluations might want to look into these hypotheses.

Besides supporting Plessner and Haar's (2006) social-cognitive perspective on sports performance judgments, the Brunswikian perspective of sports officiating ([Plessner et al., 2009](#)), and the *EASI model* ([Van Kleef, 2009](#)), the present findings add to the growing literature on nonverbal behavior in sports (see [Furley & Schweizer, 2020](#), for a recent review). Research on nonverbal behavior in sports has shown that athletes ([Furley & Schweizer, 2014a](#)) and officials ([Furley & Schweizer, 2016b](#)) automatically display certain nonverbal behaviors depending on their internal states (e.g., emotions) and observers can draw accurate inferences from these observations. Further research has shown that nonverbal behavior changes occurring during sports competitions affect observers (e.g., prospective confidence levels of opponent athletes; [Furley & Schweizer, 2014b](#)). Certain nonverbal displays have also been shown to affect the subsequent interaction with observers of these displays ([Furley, Dicks, Stendtkke, & Memmert, 2012](#); [Jordet & Hartman, 2008](#); [Moll et al., 2010](#)). The present research extends these findings by showing that postperformance nonverbal expressions can affect the evaluation of the observed performance.

The present research approach has some notable strengths and weaknesses. We consider it a strength that our stimulus material shows naturally occurring nonverbal emotional expressions of surfers during surf competitions, instead of being artificially created, which is a common limitation in research on nonverbal behavior in sports ([Buscombe, Greenlees, Holder, Thelwell, & Rimmer, 2006](#); [Furley & Dicks, 2012](#); [Furley, Dicks, & Memmert, 2012](#); [Furley et al., 2012, 2015](#); [Greenlees, Bradley, Holder, & Thelwell, 2005](#); [Greenlees, Buscombe, Thelwell, Holder, & Rimmer, 2005](#)). The multistudy nature of the present research can also be considered a strength and demonstrates the robustness of the findings across two population groups (laypeople and surf judges) and in two sets of stimuli. Therefore, it seems likely that the study's findings generalize toward judging in actual surf events. Furthermore, both studies were adequately powered to detect medium effects (equivalent of $f = 0.25$), which is an important prerequisite for increasing the reproducibility of studies (e.g., [Schweizer & Furley, 2016](#)).

The main limitation of our research is that it is not clear if the results from our online studies transfer to actual judging during surf contests. Although judging courses in surfing make use of video footage from actual competitions for training purposes, our experimental setting of watching consecutive wave sequences without the context of a heat dynamic unfolding might not be representative of real-world surf judging. For example, it is possible that the head judge during a particular surfing event might give input and point out to the individual judges to not fall for any claims of a particular surfer. Further research addressing the topic (e.g., monitoring judges during a contest) would help to scrutinize the present findings. In addition, it would be interesting to test the effect of postperformance emotional expressions in different performance contexts.

In conclusion, the present research highlights the potential influence of emotional nonverbal expressions of surfers (claims)

on observers of surf contests. More specifically, claims after successful surfing performances resulted in more favorable performance evaluations as identical performances that did not show the claims.

Note

1. We decided to use an intuitive continuous scale ranging from 0 to 100 in Experiment 1 instead of the actual 10-point scale (Experiment 2) surf judges use during competition, because participants in Experiment 1 would not have been familiar with the judging criteria in competitive surfing (e.g., scores from 8 to 10 points represent excellent performances). Instead, the continuous scale was selected to be sensitive to detect differences in performance evaluations as a function of whether claims were present or absent.

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